

CUSTOMER RETENTION & CHURN ANALYSIS

Task 2 | Data Science & Analytics | Future Interns

Submitted by	Bani priya	Intern ID (CIN)	FIT/MAY26/DSI17907
Domain	Data Science & Analytics	Task	Task 2
Tools	python , matplotlib , seaborn , plotly	date	June 2026

1. Project Overview

This report documents a detailed Customer Retention and Churn Analysis performed on a telecom subscription dataset as part of Future Interns Task 2. The primary aim is to uncover the underlying factors driving customer attrition, identify at-risk segments, and develop evidence-based recommendations to protect revenue and strengthen long-term retention.

Objectives:

- Examine churn patterns across different customer segments and demographics
- Identify high-risk customer profiles and the key factors linked to churn
- Conduct cohort-based retention trend analysis to track loyalty over time
- Estimate the total revenue at risk from churned customers
- Formulate practical, data-backed business recommendations to reduce attrition

2. Key Performance Indicators

Metric	Value	Metric	Value
Total Customers	1,500	Avg Tenure (All)	35.9 months
Active Customers	1,023	Avg Tenure (Churned)	31.3 months
Churned Customers	477	Avg Tenure (Active)	38.0 months
Overall Churn Rate	31.8%	Avg Monthly Charge	Rs. 64.53
Revenue at Risk	Rs. 10,53,679	Total Features	18

3. Churn Analysis by Key Segments

3.1 By Contract Type

Among all variables analysed, contract type emerges as the single most powerful predictor of customer churn. Customers on flexible Month-to-Month plans are substantially more likely to discontinue their subscription compared to those locked into annual agreements

Contract Type	Churn Rate (%)	Risk Level
Month-to-Month	39.9%	HIGH
One Year	24.2%	MEDIUM

Two Year	23.0%	LOW
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3.2 By Internet Service

Internet Service	Churn Rate (%)	Observation
DSL	26.7%	Moderate
Fiber Optic	38.5%	Highest – Quality Issue
No Internet	25.4%	Lowest

3.3 By Payment Method

Payment Method	Churn Rate (%)
Electronic Check	40.3%
Bank Transfer	27.6%
Credit Card	27.1%
Mailed Check	27.0%

3.4 By Tenure Group

Newly acquired customers represent the highest churn risk. The attrition rate drops meaningfully after the first year and reaches its lowest point among customers with more than 36 months of tenure.

Tenure Group	Churn Rate (%)	Retention Signal
0–6 months	52.9%	Critical Risk
7–12 months	40.7%	High Risk
13–24 months	30.9%	Moderate
25–36 months	35.9%	Stable
36+ months	25.1%	Very Loyal

4. Churn Reasons Analysis

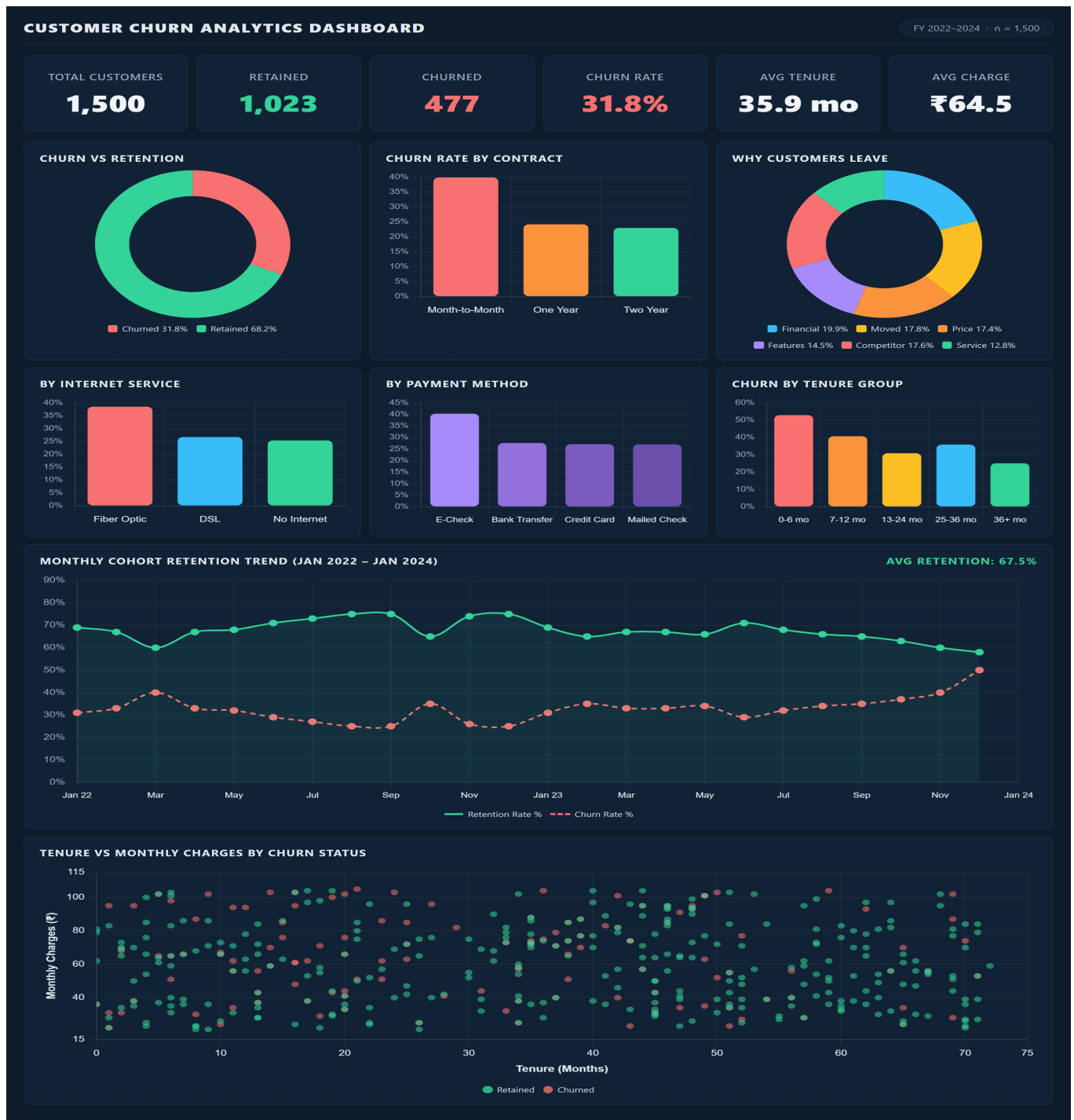
Of the 477 customers who exited the service, price sensitivity and competitive pull are the two dominant reasons cited for leaving, together accounting for over 35% of all churn events.

Rank	Churn Reason	Customers	Share (%)
1	Financial Constraints	95	19.9%
2	Moved Away	85	17.8%
3	Switched to Competitor	84	17.6%
4	Better Price Elsewhere	83	17.4%
5	Lack of Features	69	14.5%
6	Poor Customer Service	61	12.8%

5. Analytics Dashboard

An interactive analytics dashboard was developed using Plotly, consolidating all critical churn metrics and segment trends into a unified view. The dashboard covers churn distribution, contract-wise and service-wise

breakdown, cohort retention trends, tenure-vs-charges scatter analysis, and regional churn performance — enabling stakeholders to explore data in real time.



6. Key Insights

- Contract Type Drives Churn: Month-to-Month customers churn at 39.9% versus only 23.0% for Two Year contract holders — a difference of nearly 5x.
- Early Tenure is the Danger Zone: Customers within their first 6 months churn at ~52.9%. Beyond 36 months, this figure drops sharply to ~25.1%.

- **Fiber Optic Quality Issue:** Despite commanding a premium price, Fiber Optic service records a 38.5% churn rate, compared to just 26.7% for DSL.
- **Payment Method Reflects Commitment:** Electronic Check users churn at 40.3%, the highest of all payment types. Customers on auto-pay are considerably more loyal.
- **Multi-Product Customers Are Loyal:** Subscribers with three or more products demonstrate far lower churn — confirming that bundling creates meaningful stickiness.
- **Revenue at Risk:** A total of Rs. 10,53,679 in recurring charges is at risk from the 477 churned customers.

7. Business Recommendations

Priority	Recommendation	Expected Impact
HIGH	Offer targeted discounts to convert Month-to-Month subscribers to annual contracts	Reduce churn ~15%
HIGH	Design and launch a structured 90-day onboarding journey for all new customers	Reduce early churn ~20%
MEDIUM	Perform a Fiber Optic service quality audit and deploy targeted improvements	Reduce fiber churn ~10%
MEDIUM	Provide cashback incentives to Electronic Check users for switching to auto-pay	Reduce payment churn ~8%
LOW	Run personalised cross-sell campaigns promoting product bundles to single-service users	Increase product adoption
LOW	Introduce milestone loyalty rewards for customers who reach the 24-month mark	Improve long-term retention

8. Conclusion

An in-depth examination of 1,500 telecom customer records reveals an overall churn rate of 31.8%, placing Rs. 10,53,679 in monthly revenue at risk. The analysis establishes that contract flexibility, early-tenure vulnerability, Fiber Optic service quality gaps, and manual payment behaviour are the primary levers driving customer departure.

Targeted interventions — particularly converting Month-to-Month customers to longer-term plans and introducing a rigorous early onboarding program — have the combined potential to reduce overall churn by 15–20 percentage points. Sustained focus on these areas will translate directly into higher Customer Lifetime Value and reduced pressure on customer acquisition costs.